Cambridge (CIE) A Level Maths: Pure 1

Your notes

Quadratics

Contents

- Expanding Brackets
- Quadratic Graphs
- Discriminants
- Completing the square
- Solving Quadratic Equations
- Solving Hidden Quadratics using Substitutions

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Your notes

Expanding quadratics

You should be familiar with all aspects of basic algebra including manipulating expressions; simplifying; expanding and factorising brackets with both linear and quadratics. In Pure Mathematics it is important that you are confident working with quadratics.

How do I expand two brackets?

- You will have learnt a method for expanding double brackets in previous courses
  - The quickest and most common method is to use **FOIL**
    - **First** term in each bracket
    - **Outside** terms
    - **Inside** terms
    - **Last** terms
  - For $(ax + b)(cx + d)$ you will have $acx^2 + adx + bcx + bd$
- Once you have expanded and multiplied all four terms simplify the two middle terms to give you a three-term quadratic
  - $acx^2 + (ad + bc)x + bd$
- Take extra care when working with negatives
- Look out for a linear expression squared, this must still be treated as double brackets
  - $(ax + b)^2 = a^2x^2 + 2abx + b^2$
- Makes sure you can recognise when a problem involves the **difference of two squares**
  - $(ax + b)(ax - b) = a^2x^2 + abx - abx - b^2 = a^2x^2 - b^2$
  - Spotting this will be particularly useful in many areas of Pure Mathematics

How do I expand brackets with quadratic terms?

- Treat these the same as expanding double brackets with two linear expressions
- Take care with powers, remember the laws of indices
- Look carefully to see if the terms can be simplified or not
  - For $(ax^2 + b)(cx^2 + d)$ you will have $acx^4 + adx^2 + bcx^2 + bd = acx^4 + (ad + bc)x^2 + bd$
  - For $(ax^2 + b)(cx + d)$ you will have $acx^3 + adx^2 + bcx + bd$

Worked Example

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Your notes

Expand and simplify $\left(2x^2 - 4\right)(3x + 1) - (2x - 2)^2$

$(2x^2 - 4)(3x + 1) - (2x - 2)^2$ REWRITE AS DOUBLE BRACKETS

$(2x^2 - 4)(3x + 1) - (2x - 2)(2x - 2)$

$(2x^2 - 4)(3x + 1) - (2x - 2)(2x - 2)$

F O I L F O I L

USE BRACKETS FOR THE SECOND EXPRESSION

$6x^3 + 2x^2 - 12x - 4 - (4x^2 \underbrace{- 4x - 4x}_{= -8x} + 4)$

WATCH OUT FOR THE NEGATIVES

SIMPLIFY

$6x^3 + 2x^2 - 12x - 4 - 4x^2 + 8x - 4$

$6x^3 - 2x^2 - 4x - 8$

$6x^3 - 2x^2 - 4x - 8$

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Examiner Tips and Tricks

Be extra careful with negatives!

Further expanding

How do I expand brackets with more than two terms?

- To expand brackets, the rule is that each term in one set of brackets must be multiplied by each term in the other set of brackets

Your notes

$$(a+b)(x+y+z) = a(x+y+z) + b(x+y+z)$$
$$= ax + ay + az + bx + by + bz$$

$$(a+b+c)(x+y+z) = a(x+y+z) + b(x+y+z) + c(x+y+z)$$
$$= ax + ay + az + bx + by + bz + cx + cy + cz$$

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- 'FOIL' is a special case of this, when each set of brackets contains two terms
- If you are trying to expand something like $(a+b)^n$ for powers of n greater than 2 or 3, use the **binomial expansion**
- If you have to expand more than two sets of brackets, just expand them two at a time:

Worked Example

Expand $(x-4)(x+3)(x^2+x-1)$

FIRST EXPAND $(x-4)(x+3) = x^2 - x - 12$

THEN CONTINUE: $(x^2 - x - 12)(x^2 + x - 1)$

$= x^2(x^2+x-1) - x(x^2+x-1) - 12(x^2+x-1)$

$= x^4 + x^3 - x^2 - x^3 - x^2 + x - 12x^2 - 12x + 12$

$= x^4 - 14x^2 - 11x + 12$

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Worked Example

Your notes

Expand and simplify $4x(x-3y)(3x+2y-5)$.

EXPAND $4x(x-3y)$ FIRST

$4x(x-3y) = 4x^2 - 12xy$

MULTIPLY EACH TERM IN THE FIRST BRACKET BY EACH TERM IN THE SECOND BRACKET

$(4x^2 - 12xy)(3x + 2y - 5)$

$= 12x^3 + 8x^2y - 20x^2 - 36x^2y - 24xy^2 + 60xy$

$= 12x^3 - 28x^2y - 20x^2 - 24xy^2 + 60xy$

COLLECT LIKE TERMS

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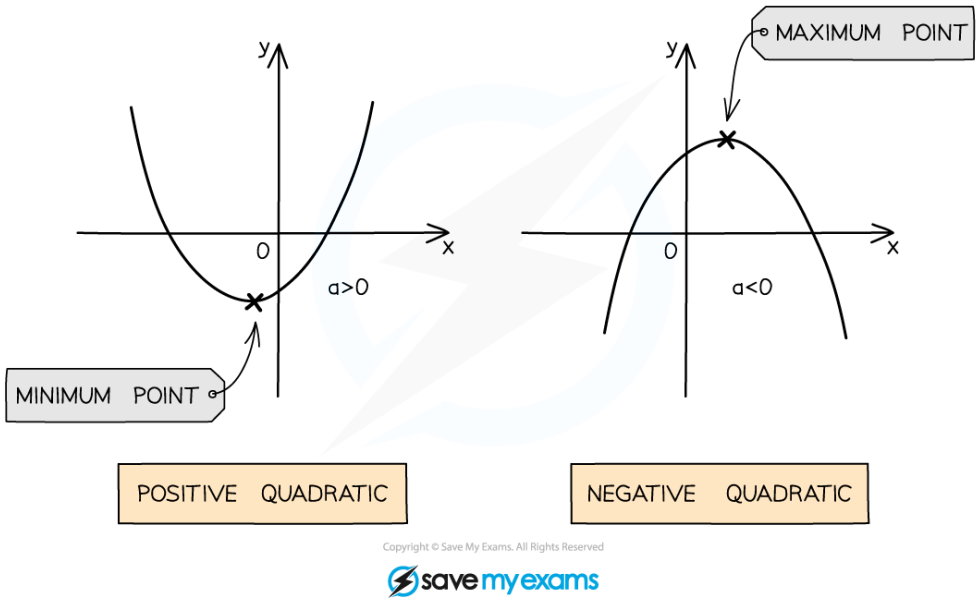
Your notes

Quadratic graphs

What are quadratic graphs?

- The general equation of a quadratic graph is $y = ax^2 + bx + c$
- Their shape is called a parabola ("U" shape)
- Positive quadratics have a value of $a > 0$ so the parabola is upright $\cup$
- Negative quadratics have a value of $a < 0$ so the parabola is upside down $\cap$



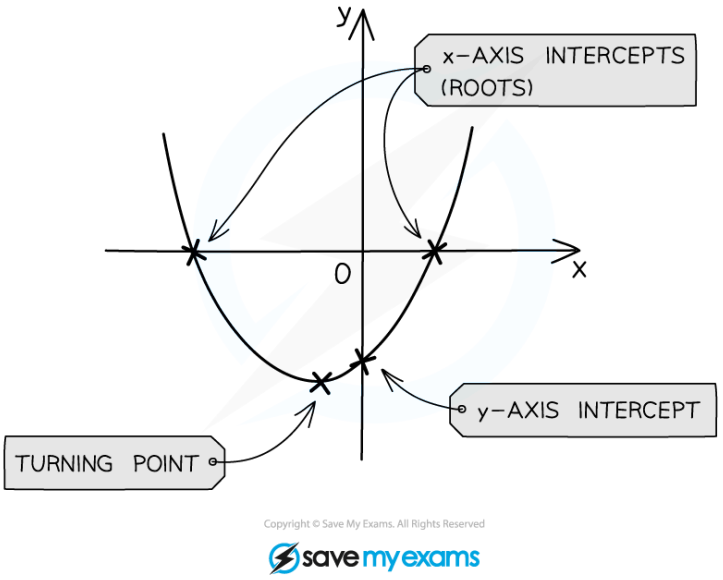
How do I use quadratic graphs?

You need to be able to:

- **sketch** a quadratic graph given an equation or information about the graph
- determine, from the equation, the **axes intercepts**
- **factorise**, if possible, to find the roots of the quadratic function
- find the coordinates of the **turning point** (maximum or minimum)

You may have to rearrange the equation before you can find some of these things

Your notes



Examiner Tips and Tricks

- Your calculator may tell you the **roots** of a quadratic function and the coordinates of the turning point
- But don't rely on it – think about how many marks the question is worth and how much method/working you should show
- Remember sometimes you'll need to rearrange an equation into the form $y = ax^2 + bx + c$

Worked Example

Your notes

Sketch the graph of $y = 2x^2 + 5x - 12$ stating the coordinates of the axes intercepts and the turning point.

$y = 2x^2 + 5x - 12$ — POSITIVE QUADRATIC: U-SHAPED

AT $x = 0$, $y = -12$ — SET $x=0$ FOR y-AXIS INTERCEPT

$y = (2x-3)(x+4)$ — FACTORISE AND SOLVE FOR x-AXIS INTERCEPTS

$y = 0$ AT $x = \frac{3}{2}$ AND $x = -4$

$\therefore$ AXES INTERCEPTS ARE $(0,-12)$, $(-4,0)$ AND $(\frac{3}{2},0)$

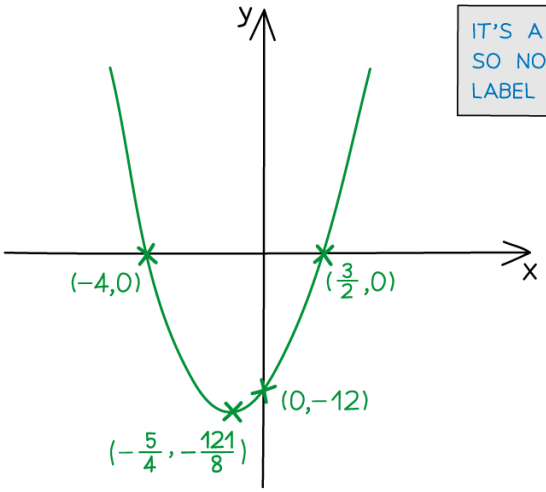
$y = 2(x^2 + \frac{5}{2}x) - 12$ — COMPLETE THE SQUARE...

$y = 2[(x+\frac{5}{4})^2 - \frac{25}{16}] - 12$

$y = 2(x+\frac{5}{4})^2 - \frac{121}{8}$ — ... TO FIND THE TURNING POINT

$\therefore$ TURNING POINT IS $(-\frac{5}{4}, -\frac{121}{8})$

IT'S A SKETCH SO NO NEED TO LABEL AXES



Your notes

Discriminants

What is a discriminant?

- The discriminant is the part of the quadratic formula that is under the square root sign $b^2 - 4ac$
- It is sometimes denoted by the Greek letter Δ (capital delta)

How does the discriminant relate to graphs and roots?

There are three options for the outcome of the discriminant:

- If $b^2 - 4ac > 0$ the quadratic crosses the x-axis **twice** meaning there are **two distinct real roots**
- If $b^2 - 4ac = 0$ the quadratic touches the x-axis **once** meaning there is **one real root** (also called repeated roots)
- If $b^2 - 4ac < 0$ the quadratic **does not cross** the x-axis meaning there are no real roots

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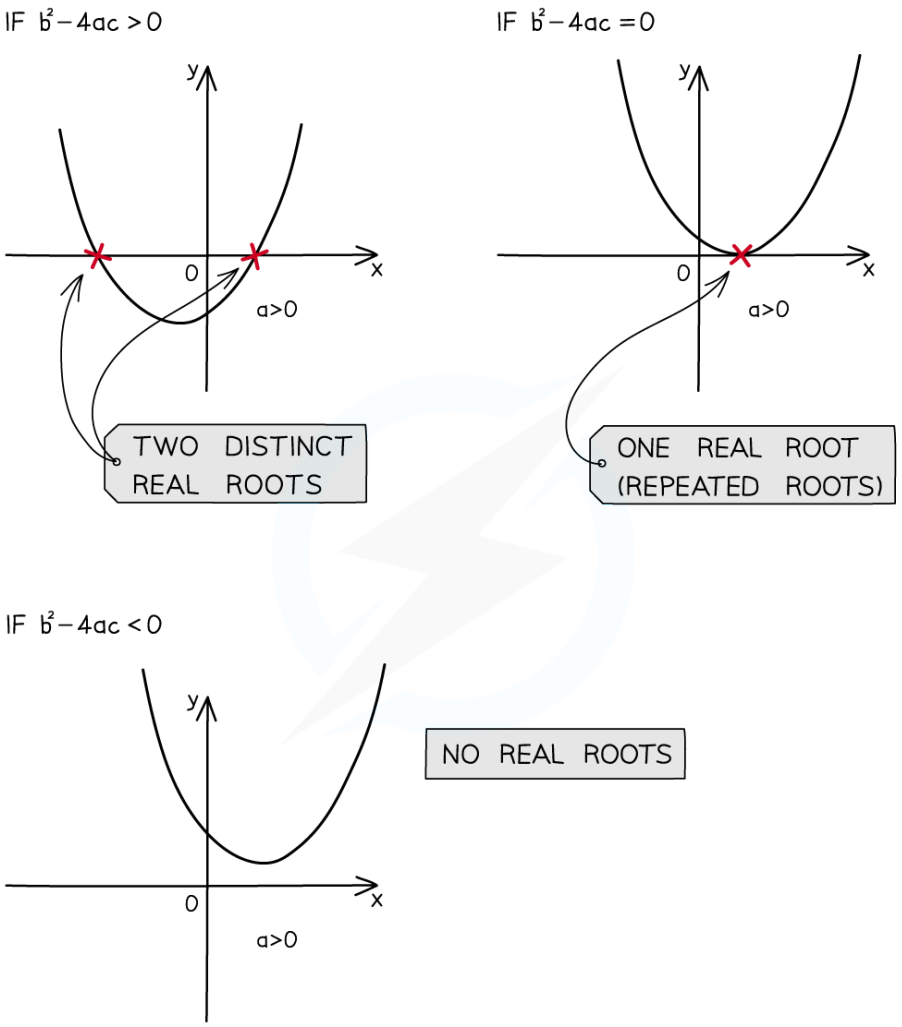

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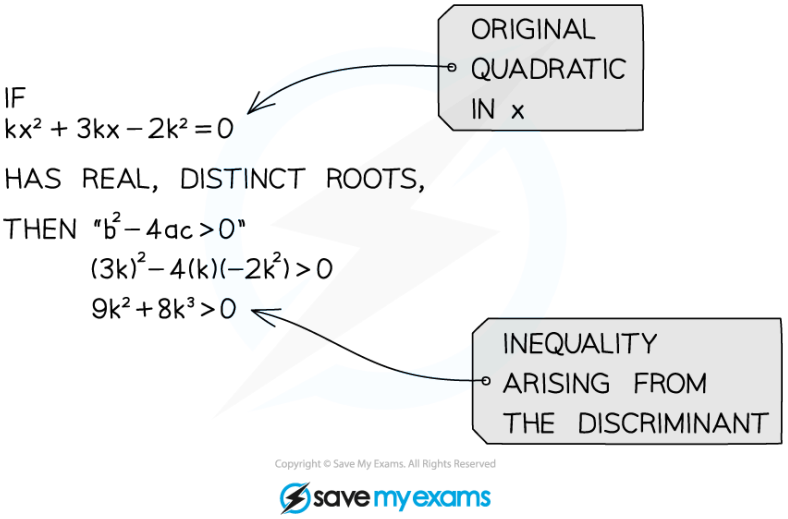
Your notes

Discriminant and inequalities

- You need to be able to **set up** and **solve** equations and inequalities (often quadratic) arising from the discriminant
- Sketch the quadratic and decide whether you're looking above or below zero to write your solutions correctly



Examiner Tips and Tricks

- When questions just mention "real roots", the roots could be **distinct** or **repeated** (i.e. they arent talking about complex numbers!)
- In these cases, you only need to worry about solving $b^2 - 4ac \geq 0$
- When solving using inequalities always sketch the quadratic and decide whether you're looking above or below zero to help write your solutions correctly

Worked Example

Your notes

Find the values of k for which the quadratic function $f(x) = 2kx^2 + kx - k + 2$ has real roots.

$f(x) = 2kx^2 + kx - k + 2$

$(k)^2 - 4(2k)(-k+2) \geqslant 0$

"$b^2 - 4ac \geqslant 0$" FOR REAL ROOTS

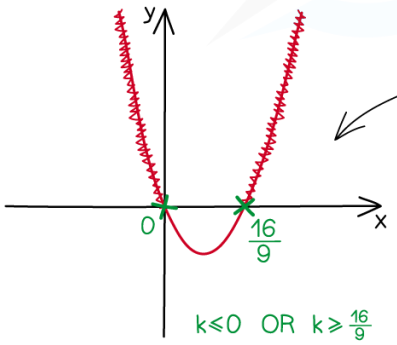
$k^2 - 8k(2-k) \geqslant 0$

$k^2 - 16k + 8k^2 \geqslant 0$

$9k^2 - 16k \geqslant 0$

$k(9k - 16) \geqslant 0$

YOUR CALCULATOR MAY SOLVE THIS...



... BUT A SKETCH IS ESSENTIAL

$k \leqslant 0$ OR $k \geqslant \frac{16}{9}$

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Your notes

Completing the square

What is completing the square?

- Completing the square is another method used to solve quadratic equations
- It simply means writing $y = ax^2 + bx + c$ in the form $y = a(x+p)^2 + q$
- It can be used to help find other information about the quadratic like coordinates of the turning point

How do I complete the square?

The method used will depend on the value of the coefficient of the **x^2** term in $y = ax^2 + bx + c$

When **a = 1**

- p is half of the coefficient of b
- q is c - p^2

e.g. $y = x^2 + 8x - 2$

STEP 1: "$p = \frac{b}{2}$" $p = \frac{8}{2} = 4$

STEP 2: "$q = c - p^2$" $q = -2 - 4^2 = -18$

STEP 3: WRITE THE ANSWER!

$y = (x+4)^2 - 18$

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When **a ≠ 1**

- You first need to take a out as a factor of the x^2 and x terms
- Then continue as above

Your notes

e.g. $y = 4x^2 + 16x + 5$

NO NEED TO INVOLVE "c"

STEP 1: FACTOR "a" ON RHS $y = 4[x^2 + 4x] + 5$

STEP 2: WORKING WITH $[x^2 + 4x]$ ONLY, "$p = \frac{b}{2}$"

$p = \frac{4}{2} = 2$

STEP 3: STILL WITH $[x^2 + 4x]$ ONLY, "$q = c - p^2$"

$q = 0 - 2^2 = -4$

STEP 4: EXPAND AND SIMPLIFY $y = 4\,[(x+2)^2 - 4] + 5$

$y = 4(x+2)^2 - 16 + 5$

$y = 4(x+2)^2 - 11$

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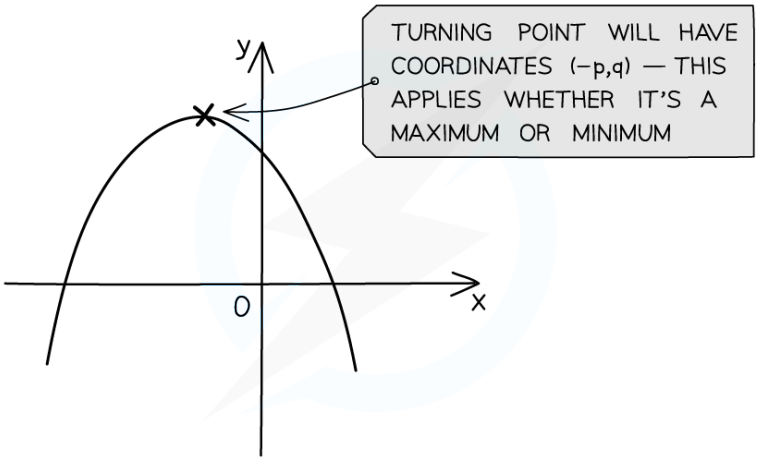
When is completing the square useful?

- Completing the square helps us find the **turning point** on a quadratic graph
- It can also help you **create the equation** of a quadratic when given the turning point



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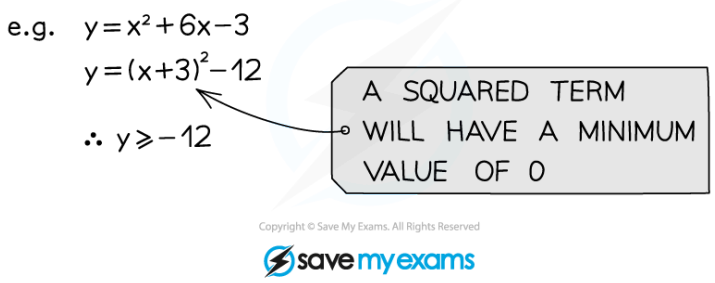
- It can also be used to **prove** and/or **show** results using the fact that a squared term will always be greater than or equal to 0



Your notes

Examiner Tips and Tricks

- Sometimes the question will explicitly ask you to complete the square
- Sometimes it will even remind you of the form to write it in
- But sometimes it will expect you to spot that completing the square is what you need to do to help with other parts of the question... like finding turning points!

Worked Example

Your notes

(a) Show $2x^2 + 6x + k > 0$ for $k > \frac{9}{2}$

(b) What can you say about the roots of the equation

$2x^2 + 6x + k = 0$ when $k = \frac{9}{2}$?

(a) ">0" IMPLIES A MINIMUM VALUE — SO COMPLETE THE SQUARE WITH "a≠1"

$2x^2 + 6x + k = 2[x^2 + 3x] + k$

STEP 1: FACTOR "a"

$= 2[(x + \frac{3}{2})^2 - \frac{9}{4}] + k$

STEP 2: "$p = \frac{b}{2}$"

STEP 3: "$q = c - p^2$"

NOTE WITH STEP 2 & STEP 3 YOU ARE ONLY WORKING WITHIN THE SQUARE [ ] BRACKETS

$= 2(x + \frac{3}{2})^2 - \frac{9}{2} + k$

$= 2(x + \frac{3}{2})^2 + k - \frac{9}{2}$

STEP 4: EXPAND AND SIMPLIFY

$2(x + \frac{3}{2})^2 \geqslant 0$ FOR ALL VALUES OF x

$\therefore 2(x + \frac{3}{2})^2 + k - \frac{9}{2} > 0$ WHEN $k - \frac{9}{2} > 0$

$k > \frac{9}{2}$

SHOW ALL STAGES IN A "SHOW THAT" QUESTION

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(b) WHEN $k=\frac{9}{2}$ THE QUADRATIC BECOMES $2(x+\frac{3}{2})^2$ AND SO THERE WILL ONLY BE ONE ROOT (OR REPEATED ROOTS) TO THE EQUATION $2x^2+6x+k=0$

THE EQUATION IS NOW $2(x+\frac{3}{2})^2=0$

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Your notes

Your notes

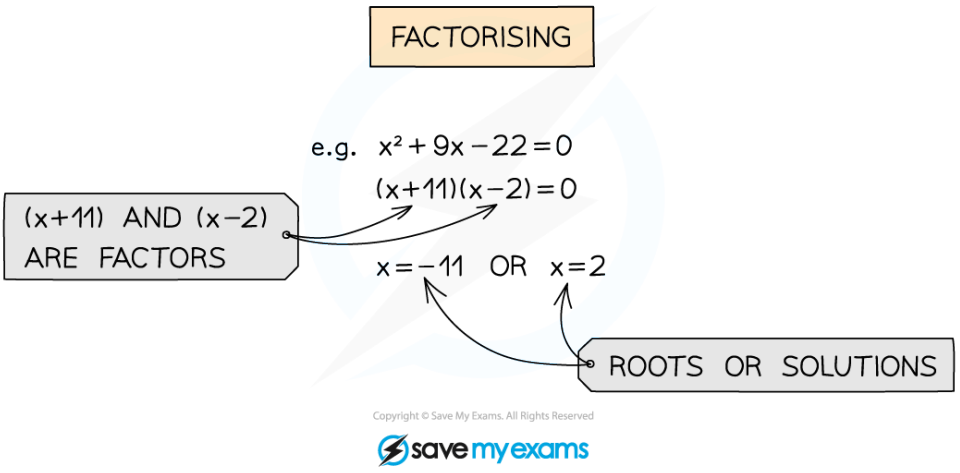
Solving quadratic equations

Solving quadratic equations

- We can solve quadratic equations when they are written in the form $ax^2 + bx + c = 0$
- If given an unusual looking equation, try to rearrange it into this form first
- The three ways to solve a quadratic you must know are
  - Factorising
  - Completing the square
  - Quadratic formula

How do I solve by factorisation?

- Factorising is a great way to solve a quadratic quickly but won't work for all quadratics
- If the numbers are simple, try factorising first
- Once factorised, set each bracket to = 0 and solve



How do I solve by completing the square?

- Completing the square will work for any quadratic
- Make sure you know how to complete the square
- Remember this will help with questions involving turning points too

Your notes

COMPLETING THE SQUARE

e.g. $x^2+10x-24=0$

$(x+5)^2-49=0$

$(x+5)^2=49$

$x+5=\pm 7$

$x=7-5$ OR $x=-7-5$

$x=2$ OR $x=-12$

COMPLETE THE SQUARE: "$p=\frac{b}{2}$", "$q=c-p^2$"

REMEMBER BOTH POSITIVE AND NEGATIVE SQUARE ROOTS

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How do I solve using the quadratic formula?

- The quadratic formula might look complicated but it just uses the coefficients a, b and c from the quadratic equation
- The quadratic formula will work for any quadratic

QUADRATIC FORMULA

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

e.g. $x^2-3x-6=0$

"b^2-4ac": $(-3)^2-4(1)(-6)=33$

$x=\frac{3\pm\sqrt{33}}{2}$

$x=\frac{3+\sqrt{33}}{2}$ OR $x=\frac{3-\sqrt{33}}{2}$

DISCRIMINANT FIRST...

... MAKES THIS LINE EASIER!

KEEP ANSWERS EXACT UNLESS INSTRUCTED OTHERWISE

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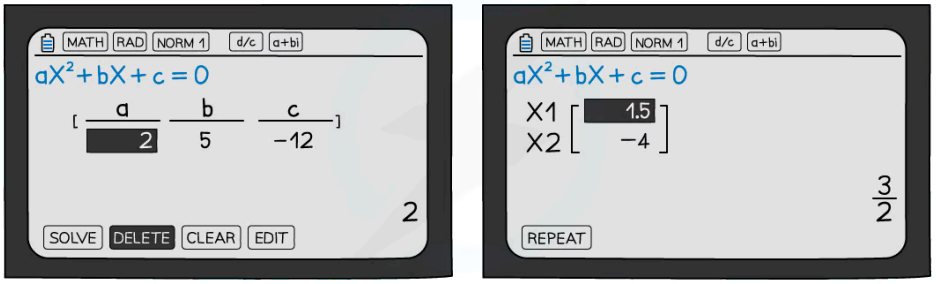
How do I solve using a calculator?

- Most calculators now have the ability to solve quadratics
- Get used to how your calculator functions work

Your notes



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- So the solution to the quadratic equation $2x^2 + 5x - 12 = 0$ are **x = 1.5** and **x = -4**

Examiner Tips and Tricks

- A calculator can be super-efficient but be aware some marks are for method
- There will never be many marks for solving a quadratic at AS/A level
- Use your judgement:
  - is it a "show that" or "prove" question?
  - how many marks?
  - how long is the question?
- Remember the quadratic formula with a song... there are loads of fun ones on YouTube

Worked Example

Your notes

Solve the equation $x + 5 = \frac{12}{x}$.

$x + 5 = \frac{12}{x}$

IT'S UNUSUAL SO TRY REARRANGING.
$\times x$ TO GET RID OF FRACTIONS

$x^2 + 5x = 12$

$x^2 + 5x - 12 = 0$

THIS DOESN'T FACTORISE, COMPLETING THE SQUARE IS NOT "NICE" DUE TO THE 5

WORKING OUT THE DISCRIMINANT FIRST MAKES THINGS EASIER AND TIDY

$"b^2 - 4ac" = 5^2 - 4 \times 1 \times (-12)$
$= 25 + 48$
$= 73$

$\therefore x = \frac{-5 \pm \sqrt{73}}{2}$

$x = \frac{-5 + \sqrt{73}}{2}$ $\quad x = \frac{-5 - \sqrt{73}}{2}$

CHECK THESE ON YOUR CALCULATOR

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Your notes

Solving hidden quadratics using substitutions

What are hidden quadratic equations?

- Hidden quadratic equations are quadratics written in terms of a function $f(x)$
- A normal quadratic appears in the form $ax^2 + bx + c = 0$
- Whereas a hidden quadratic appears in the form $a[f(x)]^2 + b[f(x)] + c = 0$
- This might look complicated but it simply means x has been replaced by $f(x)$
- e.g. $\sin^2 x + 2\sin x - 3 = 0$ is just the hidden quadratic of $x^2 + 2x - 3 = 0$ where $f(x) = \sin x$

How do I solve hidden quadratic equations?

- First rearrange the function into the form $a[f(x)]^2 + b[f(x)] + c = 0$
- Replacing the function and solving the 'normal' quadratic first
- Then substitute the function back into the solutions to solve the original quadratic

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e.g. $2\cos x = 3 - \cos^2 x \qquad 0^\circ \leqslant x \leqslant 360^\circ$

STEP 1: REARRANGE INTO THE FORM
$a[f(x)]^2 + bf(x) + c = 0$

$\cos^2 x + 2\cos x - 3 = 0$

STEP 2:
- "IGNORE" THE FUNCTION OF x AND SOLVE AS A NORMAL QUDRATIC
- USE A SUBSTITUTION, $y = f(x)$ IF YOU PREFER

$y = \cos x, \quad y^2 + 2y - 3 = 0$
$(y+3)(y-1) = 0$
$y = -3 \quad y = 1$

STEP 3: USE THESE SOLUTIONS, AND THE FUNCTION OF x, TO SOLVE THE ORIGINAL QUESTION

$\cancel{\cos x = -3}$ $\qquad \cos x = 1$
NO SOLUTIONS $\qquad x = 0^\circ, 360^\circ$

Worked Example

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Solve the equation $3^{2x+1} = 82 \times 3^x - 27$.

$3^{2x+1} - 82 \times 3^x + 27 = 0$

REARRANGE SO IT "= 0" AND WORK TOWARDS CORRECT FORM

$3^1 \times 3^{2x} - 82 \times 3^x + 27 = 0$

AWKWARD TO SPOT BUT 82×3^x IS THE CLUE

$3(3^x)^2 - 82(3^x) + 27 = 0$

THIS IS NOW IN THE FORM $a[f(x)]^2 + bf(x) + c = 0$

LET $y = 3^x$

$3y^2 - 82y + 27 = 0$

$(3y-1)(y-27) = 0$

$y = \frac{1}{3} \qquad y = 27$

YOU DO NOT HAVE TO USE A SUBSTITUTION

$\therefore\ 3^x = \frac{1}{3} \qquad 3^x = 27$

$x = -1 \qquad x = 3$

MAKE SURE YOU FIND x's FOR YOUR FINAL ANSWER

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